

Ryan M. Kim

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EDUCATION

Northeastern University

Boston, MA

B.S. in Electrical and Computer Engineering, Dean's List

Expected 2028

- Coursework: Digital Design and Computer Organization, Embedded Design: Enabling Robotics, Discrete Structures, Circuits and Signals, Fundamentals of Algorithms, Fundamentals of Networks, Physics I/II, Calculus III, Differential Equations and Linear Algebra

TECHNICAL SKILLS

Programming: Python, C/C++, Racket, Bash, MATLAB, PyTorch, scikit-learn, NumPy, Pandas

Hardware: Parallel Programming, Vivado, Quartus Prime, SystemVerilog, Verilog, LTspice

Tools: Git, Linux, L^AT_EX, Markdown, Oscilloscope, Multimeter

RESEARCH EXPERIENCE

Undergraduate Researcher

Jan. 2026 – Present

Northeastern University, under Professor Yanzhi Wang

Boston, MA

- Researching world models for robotic manipulation tasks

Research Assistant, Schiffer Lab

Aug. 2022 – Sep. 2024

UMass Chan Medical School, Dept. of Biochemistry and Molecular Biotechnology

Worcester, MA

- Investigated structural basis of influenza virus entry using cryo-electron tomography and deep learning
- Developed and trained convolutional neural networks to automatically segment viral protein components, processing 100+ tomograms and enabling subnanometer structural determination of 12,000+ particles at 8–10 Å resolution
- Contributed to quantitative morphology analysis pipeline using scikit-learn, Open3D, and EMAN2.99

PUBLICATION

Q.J. Huang, **R. Kim**, K. Song, N. Grigorieff, J.B. Munro, C.A. Schiffer, & M. Somasundaran. “Virion-associated influenza hemagglutinin clusters upon sialic acid binding visualized by cryoelectron tomography.” *Proc. Natl. Acad. Sci. U.S.A.* 122(16): e2426427122 (2025). doi.org/10.1073/pnas.2426427122

PROJECTS

Parallel Tiling Image Filters | C++, POSIX Threads

2026

- Built a multithreaded image processing tool that splits images into tiles and applies convolution-based filters using a thread pool architecture
- Implemented grayscaling, thresholding, Gaussian blur, edge detection, and sharpening
- Implemented thread-safe task scheduling with POSIX mutexes and condition variables

WM-811K Wafer Defect Analysis Pipeline | Python, Pandas, NumPy, SciPy

2025

- Developed a semiconductor defect analysis pipeline in Python to process 172,000 wafer maps across 8 classes, utilizing SciPy for image interpolation and resizing to generate averaged probability heatmaps
- Engineered an automated reporting system using Matplotlib and fpdf2 to visualize statistical trends and generate a 20-page defect “field guide”

ENGAGEMENT EXPERIENCE

Youth For Community Improvement

2021 – 2023

Greater Worcester Community Foundation

Worcester, MA

- Evaluated grant applications and allocated \$40,000 in funding to nonprofit organizations addressing community issues through a youth-led program, facilitating meetings with community speakers and authoring advocacy briefs to inform grant decision-making

Engineering Team Member, FOCUS Program

Apr. 2023 – June 2023

BAE Systems

Nashua, NH

- Designed, built, and programmed a VEX-based robot for a competitive team capstone project, applying iterative testing and engineering fundamentals to optimize performance and presenting final outcomes to BAE Systems engineering staff